

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 11 Mathematics Extension 1

Preliminary HSC Course

Assessment 2

July, 2015

Time allowed: 90 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Section I Multiple Choice
Questions 1-5
5 Marks

Section II Questions 6-11
60 Marks

SECTION I

Choose the most appropriate answer from the choices, and fill in the circle on the multiple-choice answer sheet provided in your answer booklet

1 The gradient of the tangent to the curve $y = 5x - x^3 - 2$ at the point $(2, 0)$ is

- A. 6 B. 2 C. -2 D. -7

2 $\lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x - 2} =$

- A. 0 B. -2 C. 6 D. ∞

3 The acute angle between the lines $x = 3$ and $3x - 2y - 5 = 0$, to the nearest degree, is:

- A. 56° B. 124° C. 34° D. 144°

4 If $y = 5t$ and $x = t^2$ then $\frac{dy}{dx} =$

- A. 5 B. $2t$ C. $\frac{5}{2t}$ D. $\frac{1}{2t}$

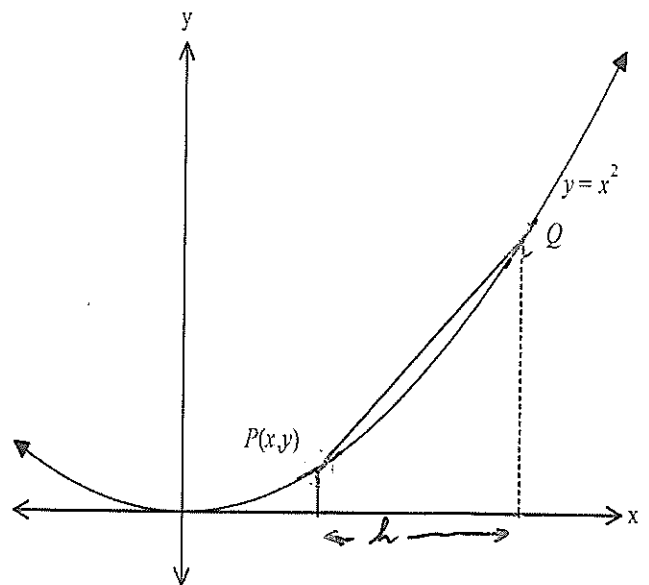
5 In the diagram at right, $f(x) = x^2$

P is the point (x, y) on the curve

Q is another point on the curve which has an x value of $x + h$

The slope of the secant PQ is given by

- A. $\frac{(x+h)^2 - x^2}{x+h}$
 B. $\frac{(x+h)^2}{h}$
 C. $\frac{h^2 - x^2}{h}$
 D. $2x + h$



SECTION II

Start each new question on a new page

QUESTION 6: (10 Marks)

Marks

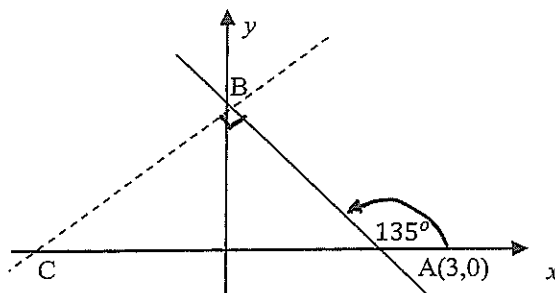
- (a) . If $0^\circ \leq \theta \leq 90^\circ$ and $\cos\theta = x$ find $\sin 2\theta$ in terms of x **2**
- (b) Find the distance between the lines $2x + 3y = 6$ and $2x + 3y + 4 = 0$ as a simplified surd. **3**
- (c) The interval joining the points A (-1, 5) to B(2, -1) is divided by the point M externally in the ratio 3:2. **2**
- Find the co-ordinates of M.
- (d) Show that $\tan x = \frac{1 - \cos 2x}{\sin 2x}$ **3**

QUESTION 7: (10 Marks) (Start a New Page)

	Marks
(a) Find derivatives of	
(i) $y = \frac{3}{x}$	1
(ii) $y = 5\sqrt{x}$	1
(iii) $y = (2x^3 - 1)(x^2 + 1)^3$ (give the answer in fully factored form)	2

- (b) A is the point (3, 0) and B is on the y axis.
AB makes an angle of 135° with the positive x-axis.

BC is drawn perpendicular to AB.

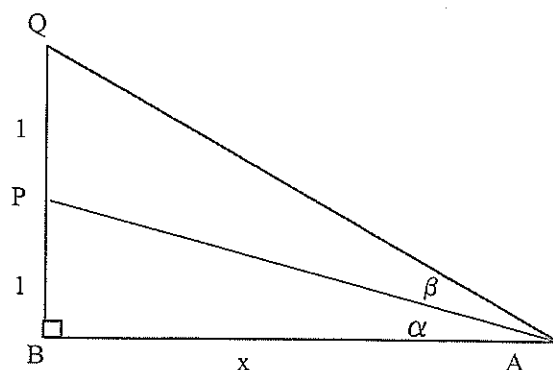


- | | | |
|------|--|---|
| (i) | Find the equation of the line BC | 2 |
| (ii) | You are further given that C lies on the (negative) x-axis. Find the area of $\triangle ABC$ | 1 |

- (c) In the diagram at right, $\triangle ABQ$ is right-angled at B. 3

$\angle QAP = \beta$ and $\angle PAB = \alpha$
 $PQ = PB = 1$ unit
 $AB = x$ units

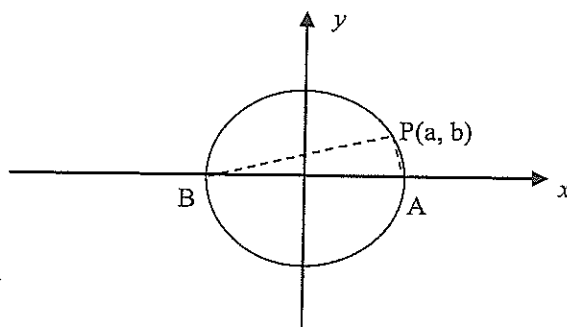
Prove that $\tan \beta = \frac{x}{x^2 + 2}$



QUESTION 8: (10 Marks) (Start a New Page)

Marks

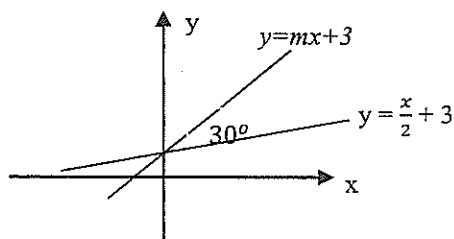
- (a) Show that the gradient of the tangent to the curve $y = \frac{x^3}{1+x^2}$ is always positive except at the origin. 2
- (b) (i) Find the exact value of $\tan 15^\circ$ in simplified form 2
- (ii) Hence find the value of $\cot 15^\circ + \tan 15^\circ$ 1
- (c) Solve the equation $2\sin^2 x + \cos x - 2 = 0$, for $0^\circ \leq x \leq 360^\circ$ 3
- (d) For the circle $x^2 + y^2 = 16$, A and B are the points where the graph cuts the x -axis. 2
- P (a, b) is a point on the circle



- (i) Find an expression for the gradient of PA
- (ii) Hence prove that $\angle APB = 90^\circ$

QUESTION 9: (10 Marks) *(Start a New Page)*

- | | | Marks |
|-----|--|--------------|
| (a) | (i) Find the value of $f'(8)$ if $f(x) = \frac{2}{\sqrt{x-4}}$ | 2 |
| | (ii) Hence, find the equation of the normal to the curve $y = f(x)$ at the point on it where $x = 8$ | 2 |
| (b) | Using the method of Differentiation from First Principles, find $\frac{dy}{dx}$ if $y = x^2 + x$ | 3 |
| (c) | The angle between the 2 lines shown below is 30° . | 3 |



Show that m has the value $\frac{18+5\sqrt{3}}{11}$

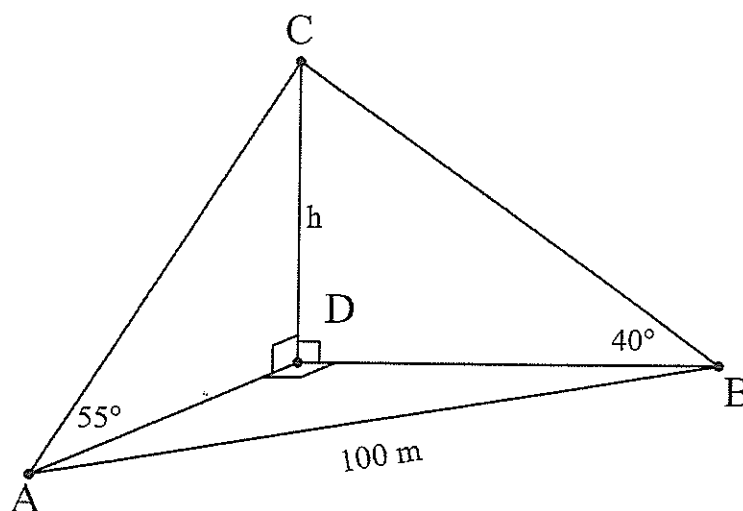
QUESTION 10: (10 Marks) *(Start a New Page)*

Marks

- (a) (i) Express $\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$,
where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$ 2
- (ii) Hence, or otherwise, solve $\cos\theta - \sin\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ 2
- (iii) What is the maximum value that $\cos\theta - \sin\theta$ can take be?
Explain your answer. 1

- (b) Two men, A and B, are standing on level ground at points 100 metres apart.

From A, who is due south of a perpendicular tower, the angle of elevation to the top of the tower is 55° . B, who is due east of the tower, notes that the tower has an angle of elevation of 40°



- (i) If h is the height of the towers, prove that 3
- $$h^2 = \frac{10\,000 \tan^2 40^\circ \tan^2 55^\circ}{\tan^2 40^\circ + \tan^2 55^\circ}$$
- (ii) Find the height of the tower, to the nearest metre. 2

QUESTION 11: (10 Marks) *(Start a New Page)*

		Marks
(a)	Shade the area given by the relationship $ x \leq y $	3
(b)	(i) Prove that $\cos 3A = 4\cos^3 A - 3\cos A$	2
	(ii) Using the above, solve $4\cos^3 A - 3\cos A = 1$ for $0^\circ \leq A \leq 360^\circ$	2
(b)	AB is a diameter of the circle $(x - 2)^2 + (y - 2)^2 = 4$, where A is the closest point on the circle to the Origin (0, 0). Find an <u>unsimplified</u> expression for the x-co-ordinate of B.	3

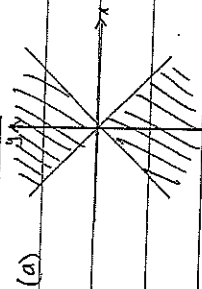
END OF EXAMINATION PAPER

SECTION I	$= \frac{\sin x}{\cos x}$
1 D	$= \tan x$
2 B	$= \text{LHS}$
3 C	<u>QUESTION 7</u>
4 C	(a) i. $y = \frac{3}{x}$
5 D	$\frac{dy}{dx} = -3x^{-2}$ $= -\frac{3}{x^2}$
SECTION II	ii. $y = 5x^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{5}{2}x^{-\frac{1}{2}}$
<u>QUESTION 6</u>	iii. $y = (2x^3 - 1)(x^2 + 1)^3$ $u = 2x^3 - 1 \quad v = (x^2 + 1)^3$ $u' = 6x^2 \quad v' = 3(x^2 + 1)^2 \cdot 2x$ $= 6x(x^2 + 1)^2$
(a)	$\frac{dy}{dx} = 6x^2(x^2 + 1)^3 + (2x^3 - 1)6x(x^2 + 1)^2$ $= 6x(x^2 + 1)^2 [x(x^2 + 1) + (2x^3 - 1)]$ $= 6x(x^2 + 1)^2 (3x^3 + x - 1)$
(b) A point on $2x + 3y = 6$ is (3, 0)	$P = \left(\frac{2 \times 3 + 3 \times 0 + 4}{\sqrt{2^2 + 3^2}} \right)$ $= \frac{10}{\sqrt{13}}$
(c)	 $A(-1, 5) \quad B(2, -1) \quad C(8, -13)$ $k_1 : k_2 = -3 : 2$ $M = \left(\frac{-3 \times 2 + 2 \times -1}{-1}, \frac{-3 \times -1 + 2 \times 5}{-1} \right)$ $= (8, -13)$
(d) RHS = $\frac{1 - \cos 2x}{\sin 2x}$	$= \frac{1 - (1 - 2 \sin^2 x)}{2 \sin x \cos x}$ $= \frac{2 \sin^2 x}{2 \sin x \cos x}$

$= \frac{1}{x}$	(c) $2 \sin^2 x + \cos x - 2 = 0$
$\frac{x^2 + 2}{x^2}$	$2(1 - \cos^2 x) + \cos x - 2 = 0$
$= \frac{1}{x} \times \frac{x^2}{x^2 + 2}$	$2 - 2 \cos^2 x + \cos x - 2 = 0$
$= \frac{x}{x^2 + 2}$	$2 \cos^2 x - \cos x = 0$
<u>QUESTION 8</u>	$\cos x (2 \cos x - 1) = 0$
(a)	$\cos x = 0 \quad \text{or} \quad \cos x = \frac{1}{2}$ $x = 90^\circ, 270^\circ \quad x = 60^\circ, 300^\circ$
	(d) A(4, 0) B(-4, 0)
	i. $m_{PA} = \frac{b}{a-4}$
	ii. $m_{PB} = \frac{b}{a+4}$
	$m_{PA} \times m_{PB} = \frac{b^2}{a^2 - 16}$
	and since $a^2 + b^2 = 16$ $a^2 - 16 = -b^2$
	$\therefore m_{PA} \times m_{PB} = \frac{b^2}{-b^2} = -1$
	$\therefore PA$ is perpendicular to PB
	<u>QUESTION 9</u>
	(a) i. $f(x) = \frac{2}{\sqrt{x-4}}$ $= 2(x-4)^{-\frac{1}{2}}$
	ii. $\cos 15^\circ + \tan 15^\circ$ $= \frac{2 - \sqrt{3}}{2} + 2 - \sqrt{3}$ $= \frac{2 + \sqrt{3}}{2} + 2 - \sqrt{3}$ $= 4$
	$f'(x) = 2 \times -\frac{1}{2} (x-4)^{-\frac{3}{2}}$ $= \frac{-1}{(x-4)^{\frac{3}{2}}}$
	$f'(8) = \frac{-1}{4^{\frac{3}{2}}} = -\frac{1}{8}$

ii. $m_{normal} = 8$ (8,1)	$\therefore \cos \theta - \sin \theta = \sqrt{2} \cos(\theta + 45^\circ)$
$y - 1 = 8(x - 8)$	
$\therefore y = 8x - 63$	ii. $\cos \theta - \sin \theta = 1$
(b) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$	$\sqrt{2} \cos(\theta + 45^\circ) = 1$
$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h}$	$\cos(\theta + 45^\circ) = \frac{1}{\sqrt{2}}$
$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h}$	$\therefore \theta + 45^\circ = 45^\circ \text{ or } 315^\circ \text{ or } 405^\circ$
$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + h}{h}$	$\therefore \theta = 0^\circ, 270^\circ, 360^\circ$
$= \lim_{h \rightarrow 0} \frac{2x + h + 1}{1}$	iii. $\cos A \leq 1$ for all A
$= 2x + 1$	$\therefore \cos \theta - \sin \theta \leq \sqrt{2}$
(c) $m_1 = m$ and $m_2 = \frac{1}{2}$	b) i. let $AD = x$ and $BD = y$
$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$	$\frac{h}{x} = \tan 55^\circ$ & $\frac{h}{y} = \tan 40^\circ$
$\frac{1}{\sqrt{3}} = \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m}$	$x = h$ $y = h$
$\sqrt{3}m = \frac{\sqrt{3}}{2} = 1 + \frac{1}{2}m$	$\tan 55^\circ$
$m(2\sqrt{3} - 1) = 2 + \sqrt{3}$	since $x^2 + y^2 = 100^2$
$m = \frac{2 + \sqrt{3}}{2\sqrt{3} - 1}$	$\frac{h^2}{\tan^2 55^\circ} + \frac{h^2}{\tan^2 40^\circ} = 100^2$
$= \frac{2\sqrt{3} - 1}{8 + 5\sqrt{3}}$	$h^2(\tan^2 40^\circ + \tan^2 55^\circ) = 100^2$
	$h^2(\tan^2 40^\circ + \tan^2 55^\circ) = 10000 \tan^2 55^\circ$
	$\therefore h^2 = \frac{10000 \tan^2 55^\circ \tan^2 40^\circ}{\tan^2 40^\circ + \tan^2 55^\circ}$
QUESTION 10	ii. $h = 72m$
(a) i. $R = \sqrt{2}$	
Using $\cos(\theta + \alpha) = \cos \theta \cos \alpha - \sin \theta \sin \alpha$	
$\cos \theta - \sin \theta = \sqrt{2}(\frac{1}{\sqrt{2}} \cos \alpha - \frac{1}{\sqrt{2}} \sin \alpha)$	
$\cos \alpha = \frac{1}{\sqrt{2}}$	
$\alpha = 45^\circ$	

QUESTION 11



(b) i. Prove $\cos 3A = 4 \cos^3 A - 3 \cos A$

$$LHS = \cos(2A + A)$$

$$= \cos 2A \cos A - \sin 2A \sin A$$

$$= (2 \cos^2 A - 1) \cos A - 2 \sin A \cos A \sin A$$

$$= 2 \cos^3 A - \cos A - 2 \sin^2 A \cos A$$

$$= 2 \cos^3 A - \cos A - 2(1 - \cos^2 A) \cos A$$

$$= 2 \cos^3 A - \cos A - 2 \cos A + 2 \cos^3 A$$

$$= 4 \cos^3 A - 3 \cos A$$

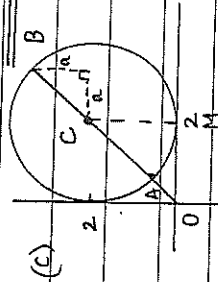
$$= RHS$$

ii. $4 \cos^3 A - 3 \cos A = 1$ $0^\circ \leq A \leq 360^\circ$

$$\cos 3A = 1 \quad 0^\circ \leq 3A \leq 1080^\circ$$

$$3A = 0^\circ, 360^\circ, 720^\circ, 1080^\circ$$

$$\therefore A = 0^\circ, 120^\circ, 240^\circ, 360^\circ$$



$$\text{since } BC = 2$$

$$\text{then } OB = 2\sqrt{2} + 2$$

O, A, C and B are in a straight line

For B:

$$x^2 + x^2 = (2\sqrt{2} + 2)^2$$

$$OC^2 = OM^2 + MC^2$$

$$= 4 + 4$$

$$= 8$$

$$\therefore OC = 2\sqrt{2}$$

$$OR \quad a^2 + a^2 = 4$$

$$2a^2 = 4$$

$$a^2 = 2$$

$$a = \sqrt{2}$$

$\therefore x$ value of B

$$is \quad 2 + \sqrt{2}$$